

Olist E-commerce Customer Segmentation using Machine Learning.

OBJECTIVES: UNCOVERING DATA-DRIVEN INSIGHTS TO OPTIMIZE MARKETING ROI & OPERATIONAL EFFICIENCY.

Presenter: Khang Ly - Data Analyst Candidate.

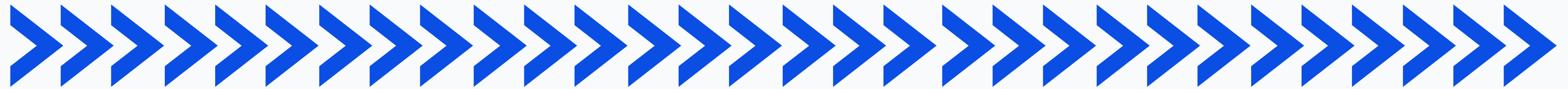


The Business Problem

- Olist has a massive customer base, but **retention rates need improvement**.
- "One-size-fits-all" marketing campaigns lead to **wasted budget**.
- **1-star reviews are frequent**, but the root causes (Shipping vs. Product Quality) are tangled and unclear.



Data Pipeline & Tech Stack



SQL

- Data extraction
- Handling nulls
- Calculating RFM
- Logistics metrics.



PYTHON

- EDA
- Log-transformation,
- Scaling
- K-Means clustering (Elbow + Silhouette).

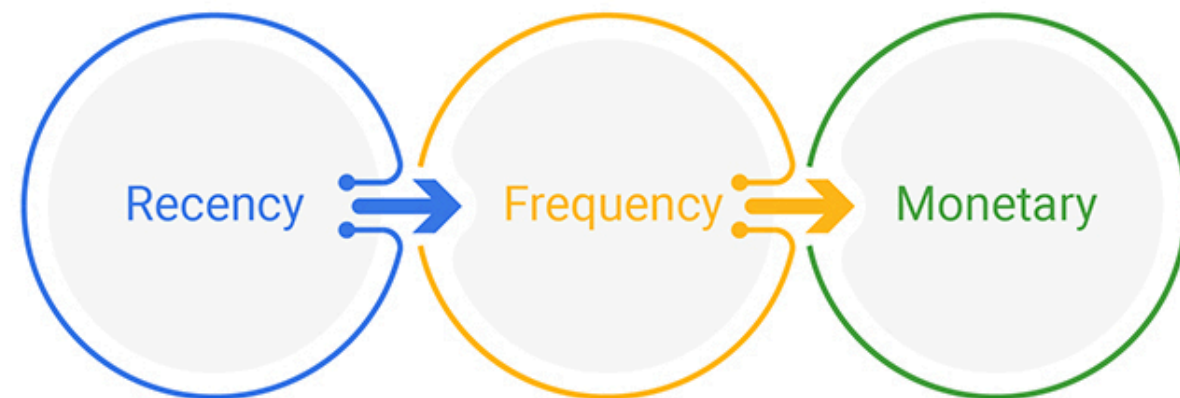


POWER BI

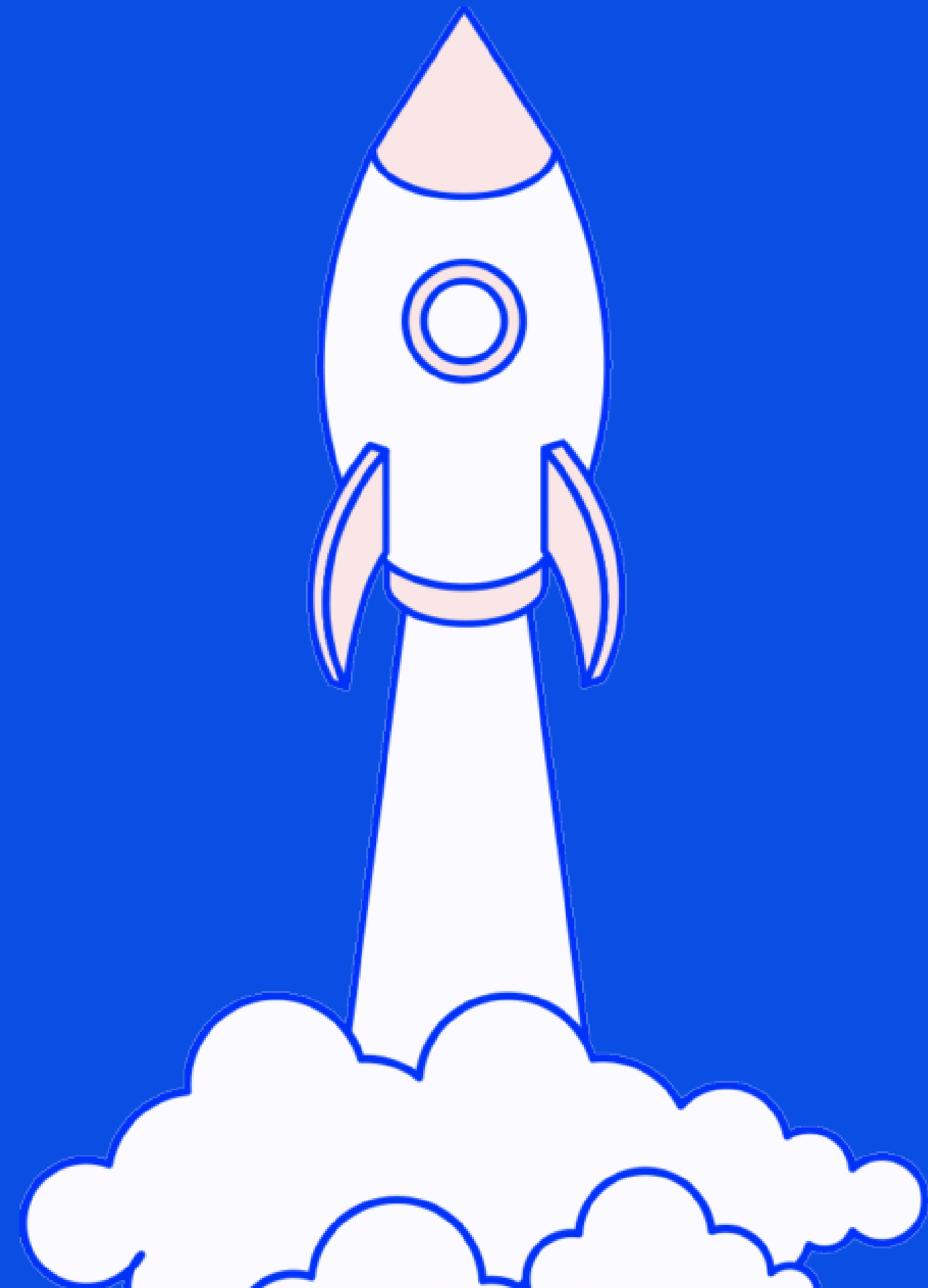
Building an automated, interactive reporting dashboard.

Feature Engineering

- **Beyond traditional RFM:** max_delay_days and avg_freight_ratio to capture the *true* customer experience.
- **Handling Skewness:** Applied Log-transformation to normalize the monetary_value distribution.



Improving Customer Segmentation With RFM Marketing Analysis



Model Selection: K-means

K = 6 provided the perfect balance between **mathematical accuracy** and **actionable business value**



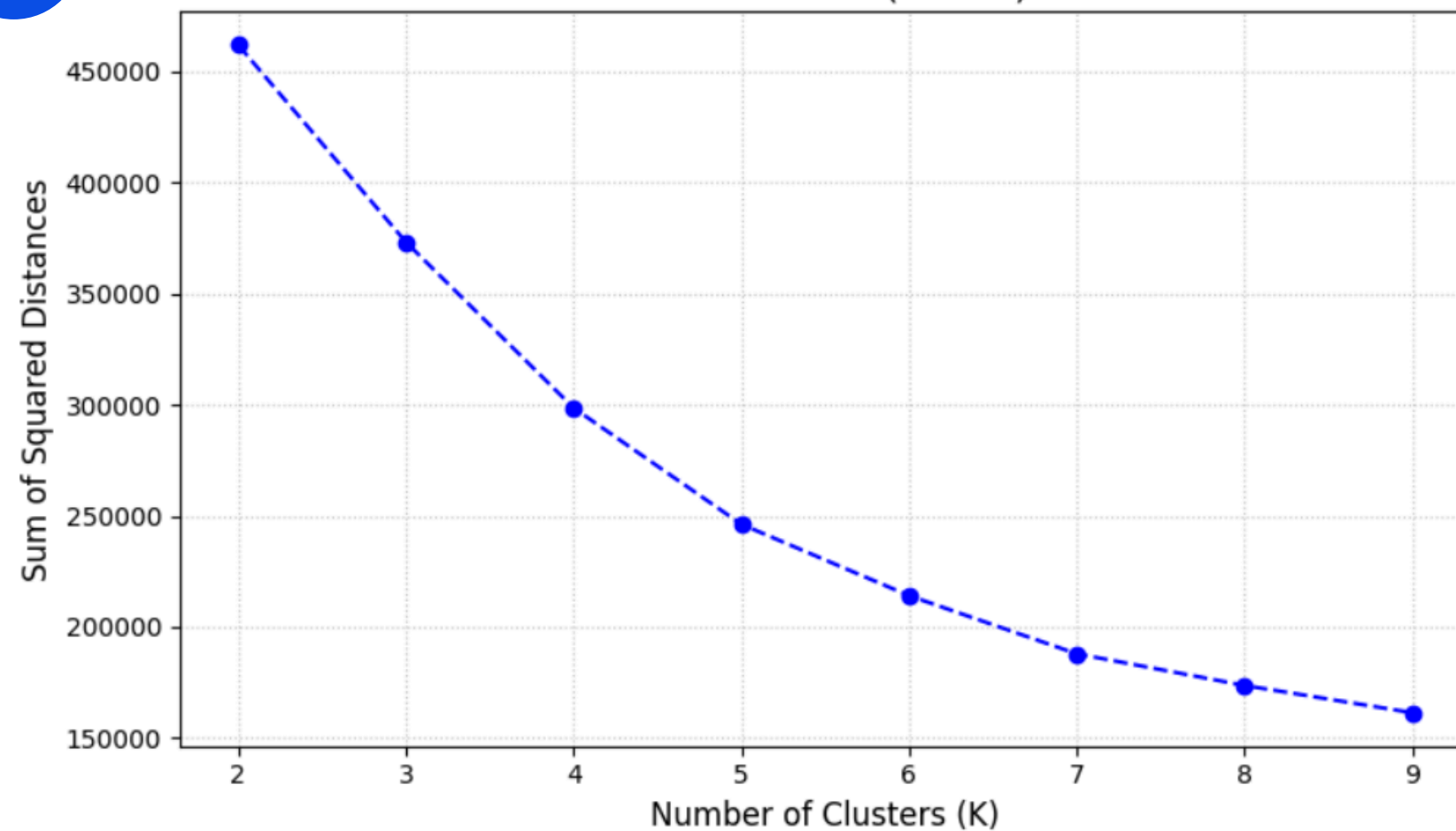
Elbow Method: identifies K by minimizing inertia (within-cluster sum of squares, WCSS), looking for an "elbow" in the plot.



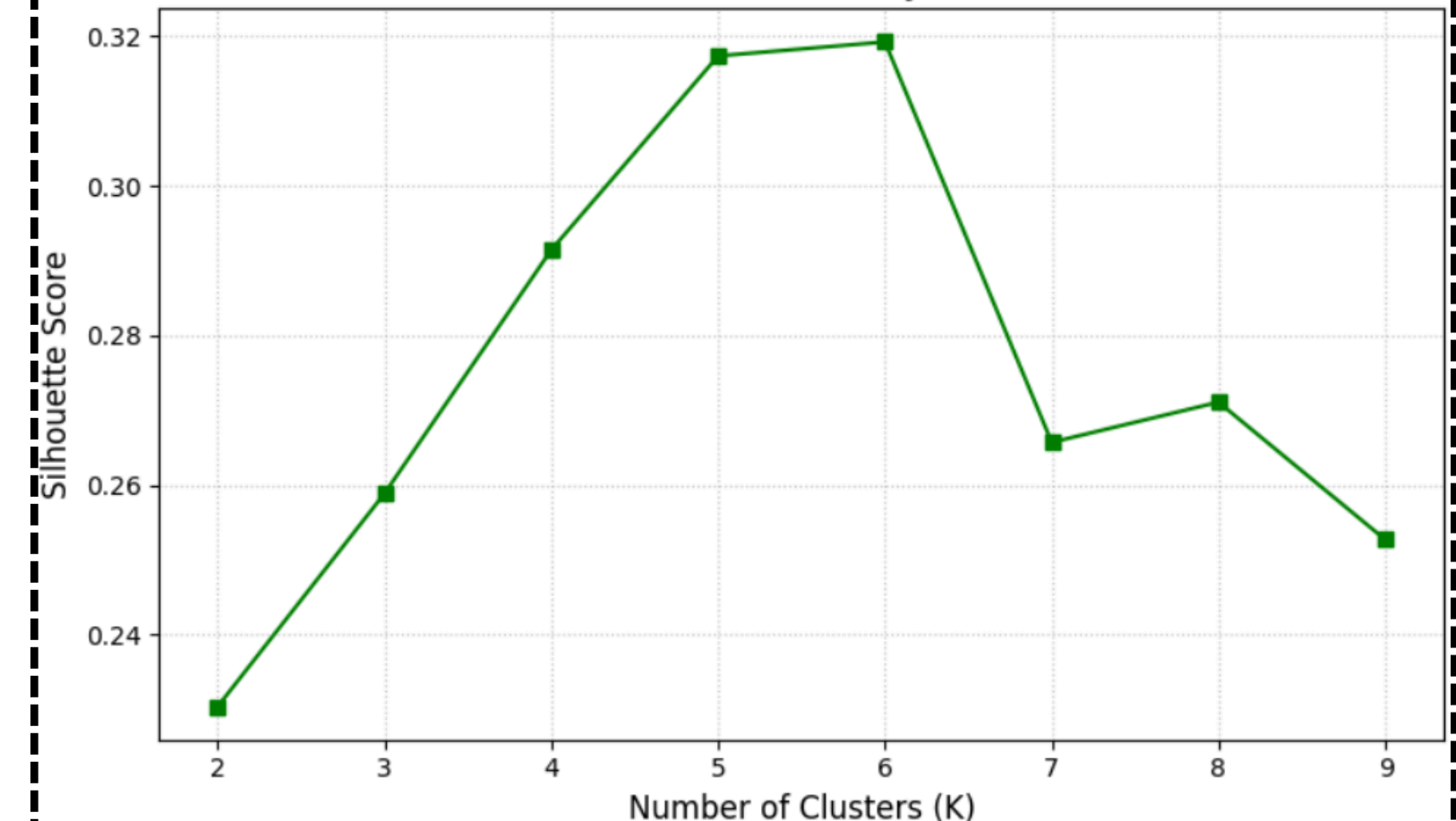
Silhouette Analysis: evaluates cluster cohesion and separation, selecting K with the highest average silhouette score [-1, +1].



Elbow Method (Inertia)



Silhouette Analysis

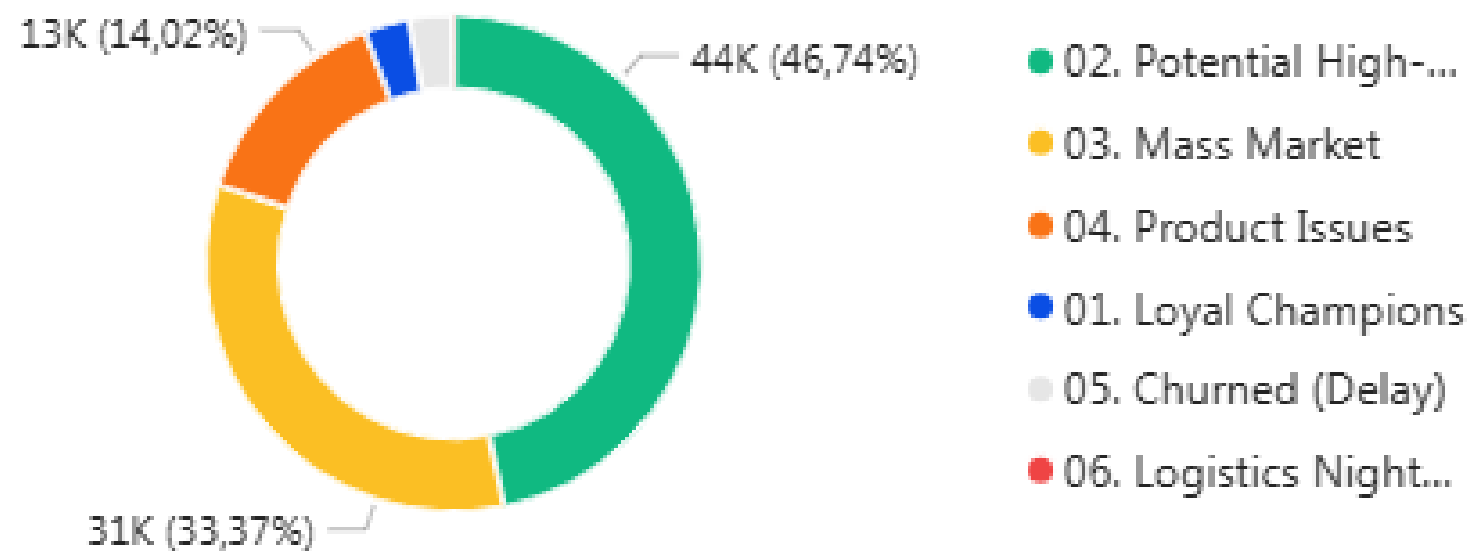


6 CUSTOMER PERSONA

The **Potential High-Value** group accounted for the **highest** proportion in terms of both number and revenue, while the **logistics-nightmare** group performed the **worst** in both numbers and revenue.



Total Customers by Cluster_Name



Revenue by Cluster_Name



6 CUSTOMER PERSONA

Divide 6 customer clusters into 2 main segments: The **High-Value Segments** and **The At-Risk Segments**



The High-Value Segments: Potential High-Value, Loyal Champions and Mass Market.



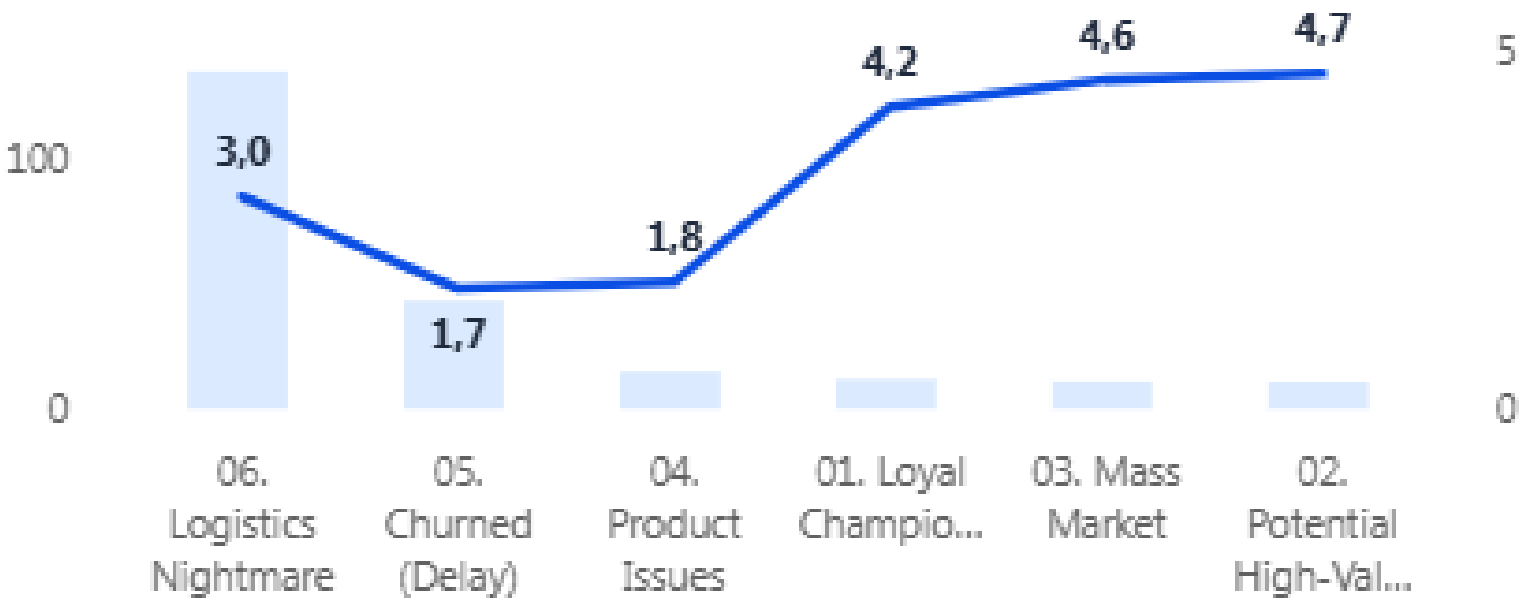
The At-Risk Segments: Logistics Nightmare, Churned (Delay) and Product Issues.



Avg order value by Cluster



Avg Delivery time vs Review score



Deep Dive - The High-Value Segments



Loyal Champions

- **Core data:** Highest purchase frequency (**2.12** times), highest value delivery (**309.41**), on-time delivery (**0.69** day delay), excellent rating (**4.23** stars).
- **Identification:** These are the "heart" of the business. They are the only customers on the platform who tend to return for repeat purchases. They spend a lot, have a smooth shipping experience, and are very satisfied with Olist.



Potential High-Value

- **Core data:** Only one purchase, but extremely high order value (**226.82**), fastest delivery on the platform (**10.8 days**, no delays), super cheap shipping fee (**13%**), highest satisfaction rating on the platform (**4.70** stars).
- **Identification:** This group's experience can be described as "perfect." They purchased high-value items but the shipping fee was very cheap compared to the item. Excellent delivery speed resulted in a flurry of 5-star ratings, but the weakness is that they haven't returned for a second purchase.



Mass Market

- **Core data:** Purchase frequency: 1 time, lowest order value (59.08), extremely high shipping cost (32% of order value), but very good rating (4.61 stars).
- **Identification:** This group is the largest. They buy small, low-value items on the platform. Because the goods are so cheap, the shipping cost is sometimes one-third of the item price (32%). Despite this, due to the fast delivery time (10.95 days), they still happily give high ratings.

Cluster	frequency	monetary_value	avg_order_value	avg_delivery_time	max_delay_days	avg_freight_ratio	customer_review_score
01. Loyal Champions	2.12	309.41	146.18	11.88	0.69	0.22	4.23
02. Potential High-Value	1.00	226.82	226.82	10.80	0.06	0.13	4.70
03. Mass Market	1.00	59.08	59.08	10.95	0.07	0.32	4.61

Deep Dive - The At-Risk Segments



Churned by Delay

- **Core data:** The order value was quite high (190.95) but the wait time was very long (43.56 days), with a delay of 17.14 days. The review score was at rock bottom (1.69 stars).
- **Identification:** The fault lies entirely with the slow shipping process. Customers spent a considerable amount of money but had to endure a delay of over half a month. This disappointment translated into devastating 1-star reviews.



Product Issues

- **Core data:** Excellent delivery time (15.05 days, delay of only 0.69 days), but a terrible rating (1.77 stars).
- **Identification:** A perfect filter of the K-Means algorithm! Logistics does its job very well, customers receive their goods on schedule. The 1-star rating here is 100% due to receiving the wrong item, counterfeit goods, damaged goods, or poor seller attitude.



Logistics Nightmare

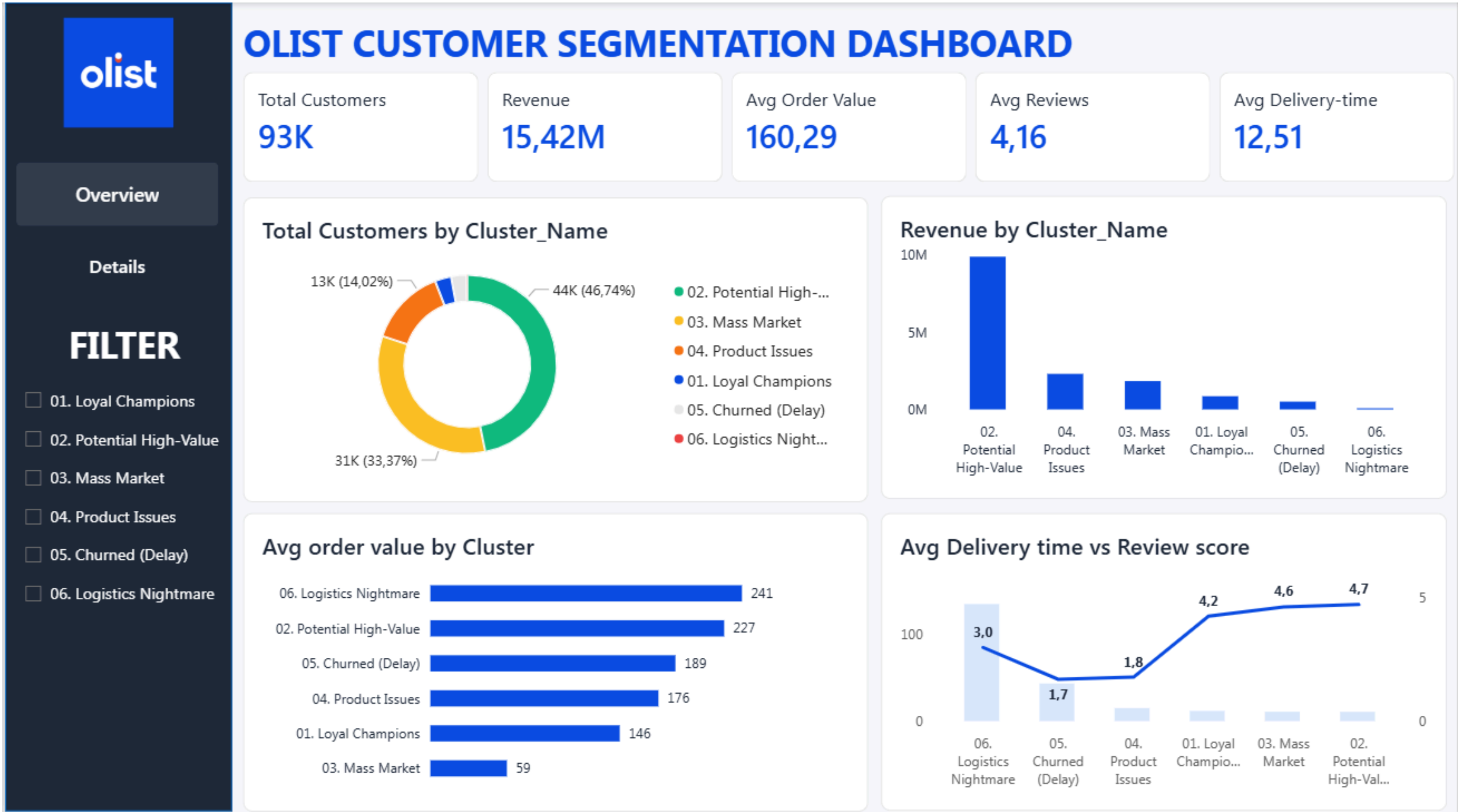
- **Core data:** Peak lead times reached 135.65 days (over 4 months), with delays of 107.81 days. Customers had high unit prices (240.53). Average rating (2.97 stars).
- **Identification:** These are "ghost orders" or orders stuck in the supply chain due to force majeure (bulky goods, incorrect address, natural disasters, lost goods at the warehouse). The 2.97 rating may be due to our data being filled in with a null value at the median, as customers waited so long that they abandoned the application and didn't bother to leave a review.

Cluster	frequency	monetary_value	avg_order_value	avg_delivery_time	max_delay_days	avg_freight_ratio	customer_review_score
04. Product Issues	1.00	176.05	176.05	15.05	0.69	0.20	1.77
05. Churned (Delay)	1.01	190.95	189.19	43.56	17.14	0.22	1.69
06. Logistics Nightmare	1.00	240.53	240.53	135.65	107.81	0.18	2.97

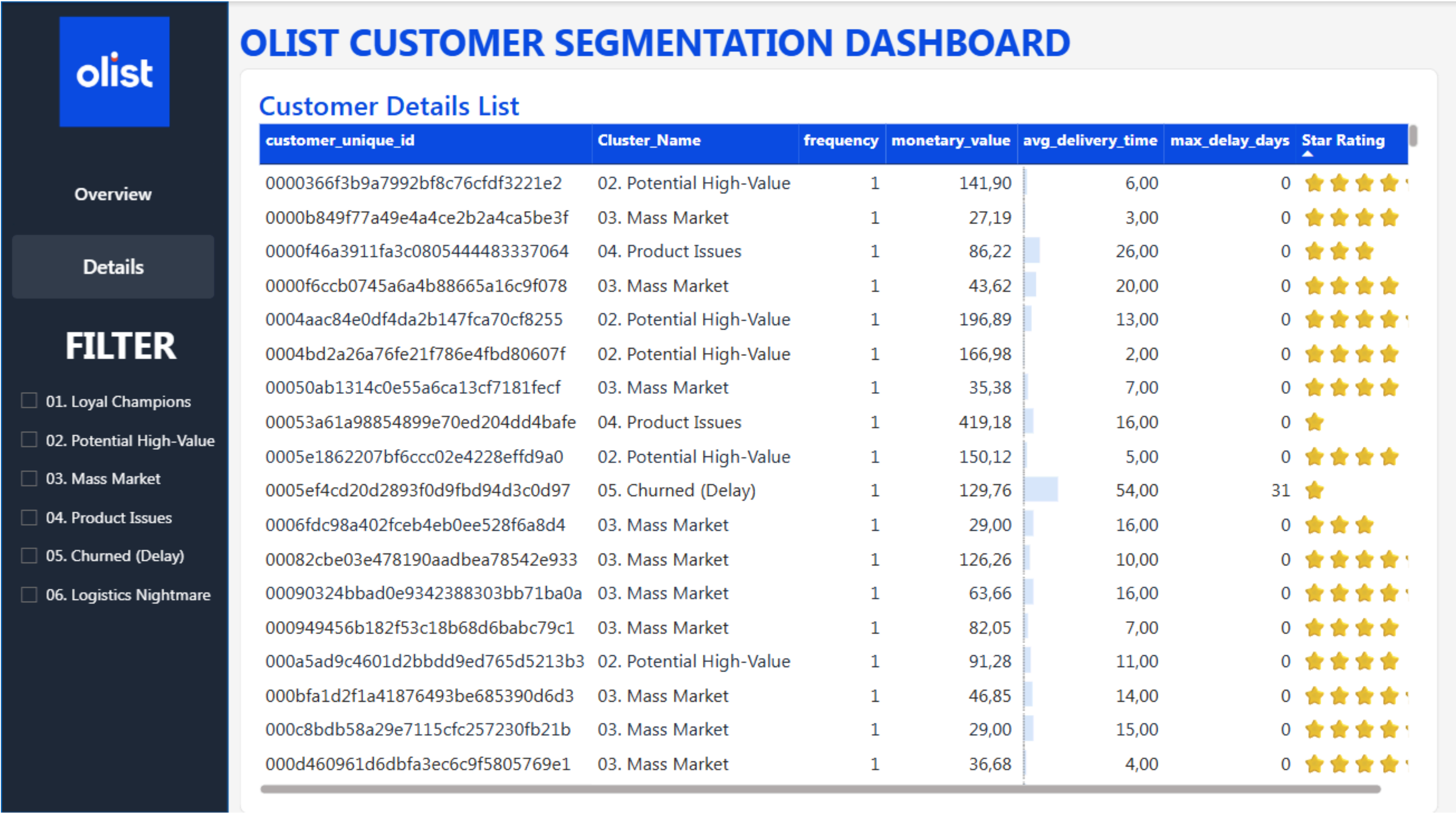
Strategic Recommendations

Segments	Strategic Actions
Loyal Champions	Immediately implement a VIP customer care program. Offer membership cards, provide lifetime free shipping codes, or priority order processing benefits to retain this "gold mine."
Potential High-value	Activate your Remarketing campaign immediately. Use Email Marketing or Push Notifications to suggest upsell/cross-sell products (accessories that complement the expensive item they just bought) along with a discount code for their second order.
Mass Market	Optimize the average order value (AOV) of this group by selling in combos (Buy 2, pay for 1; Buy an additional item to get free shipping) to reduce the feeling of "wallet pain" caused by excessively high shipping fees.
Churned by Delay	The customer service department needs to proactively call or email to apologize, clearly explain the reason for the delay, and immediately offer a compensation voucher. Operationally, a review of the delivery partners responsible for this route cluster is necessary.
Product Issues	Immediately identify the sellers supplying goods to orders in this cluster for quality verification (Audit). Simultaneously, establish a fast return/refund policy to protect buyer rights.
Logistics Nightmare	A list of these order codes needs to be immediately generated for reconciliation with third-party shipping companies (3PLs), to request compensation for damages to the platform, and to definitively resolve the processes causing the bottlenecks.

Interactive Dashboard



Interactive Dashboard





THANK YOU

- **Future Scope**

Integrating demographic data (Age, Gender, Location) for richer personas.

- **Contact**

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